

- (ii) **Heterotrophs** : Most of the bacteria are heterotrophic i.e., they can not manufacture their own food.
- They receive their own food from dead organic matter or living organism.

These are of following types :

- (a) **Saprotrophic bacteria** : These bacteria obtain food from dead and decaying organic matter.
E.g. *Clostridium botulinum*, *Lactobacillus*.
- (b) **Parasitic bacteria** : They obtain their food from living organism.
- (c) **Symbiotic bacteria** : These bacteria convert atmospheric nitrogen into nitrogenous compounds like Amino acid, NO_3 or Salts of ammonia. **E.g.** *Rhizobium*, *Frankia*.

Exercise 1.3

- Sole member of kingdom monera is :
 (1) Archaeobacteria (2) Mycoplasma
 (3) Cyanobacteria (4) Bacteria
- Which of the following have most extensive diversity in nutrition ?
 (1) Bacteria (2) Plants
 (3) Animals (4) Fungi
- All nitrifying bacteria are
 (1) Chemosynthetic heterotroph
 (2) Chemosynthetic autotroph
 (3) Chemosynthetic organotroph
 (4) Photosynthetic Lithotrophs
- Which types of bacteria are most abundant in nature ?
 (1) Heterotrophic (2) Photoautotrophic
 (3) Organotrophic (4) Symbiotic
- Genetic material of prokaryotic cell contains :
 (1) Non histonic double stranded DNA
 (2) Histonic double stranded DNA
 (3) Neither histone nor DNA
 (4) Histone without DNA
- Chemosynthetic autotrophic bacteria :
 (1) Oxidize various inorganic substance such as ammonia and use the released energy for their ATP production
 (2) Play a great role in recycling of nutrients like nitrogen, phosphorus, iron and sulphur
 (3) Both (1) and (2)
 (4) Can evolve oxygen during photosynthesis
- The function of mesosome in prokaryotes is:
 (1) Aerobic respiration
 (2) Cell wall formation
 (3) Both (1) and (2)
 (4) N_2 – fixation
- Plasmid are-
 (1) Virus
 (2) New types of micro organism
 (3) Extra chromosomal genetic material of bacteria
 (4) Essential bacterial genetic materials
- The vast majority of bacteria are:
 (1) Symbiotic (2) Autotrophic
 (3) Heterotrophic (4) Nitrogen fixing
- Infoldings of plasma membrane in bacteria are called as :-
 (1) Episomes (2) Plasmid
 (3) Pili (4) Mesosomes
- The main difference between gram +ve and gram -ve resides in the composition of
 (1) Cilia (2) Cell-wall
 (3) Cell-membrane (4) Cytoplasm
- Which of the following are the important decomposers of dead organisms?
 (1) Bacteria which are saprobic
 (2) Saprotrophic fungi
 (3) Both (1) and (2)
 (4) Parasitic plants

Reproduction :

In BGA reproduction is done by two main processes:

- (1) Vegetative (2) Asexual

(1) Vegetative Reproduction :

(i) **Binary fission** : This is the most common method of reproduction in prokaryotes. By this process only unicellular prokaryotes can reproduce. **E.g. *Spirulina***

(ii) **Fragmentation** : Filamentous prokaryotes, reproduce by this process. **E.g. *Oscillatoria*, *Nostoc***

Economic Importance of BGA :-**Useful activities :**

- (1) **Nitrogen fixation** : They provide fertility to soil by nitrogen fixation. **E.g. *Nostoc*, *Anabaena*, *Aulosira*, *Oscillatoria*** Food supplement : **E.g. *Spirulina***

Harmful activities :**(1) Water bloom :**

Excessive growth of BGA in polluted water.

BGA grow rapidly in polluted water and secrete toxic substances. **E.g. *Anabaena***

- (2) **BGA that grow in water tanks, pollute the water** : **E.g. *Oscillatoria***

Exercise 1.5

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| <p>1. Which of the following is very simple in structure but very complex in behaviour ?
 (1) Bacteria (2) Archaeobacteria
 (3) Cyanobacteria (4) Mycoplasma</p> <p>2. Cell wall of cyanobacteria similar to :
 (1) Gram negative eubacteria
 (2) Gram positive eubacteria
 (3) Archaeobacteria
 (4) All of the above</p> <p>3. Which of the following is nitrogen fixing organisms
 (1) <i>Nostoc</i> (2) <i>Anabaena</i>
 (3) <i>Oscillatoria</i> (4) All of the above</p> <p>4. Nutrition in Cyanobacteria is
 (1) Similar to Archaeobacteria
 (2) Similar to humans
 (3) Non-oxygenic photoautotrophic
 (4) Similar to plants</p> <p>5. BGA is found in :
 (1) Fresh water (2) Marine water
 (3) Terrestrial habitat (4) All of the above</p> <p>6. Which of the following is oxygenic filamentous autotrophic BGA :
 (1) <i>Anabaena</i> (2) <i>Nostoc</i>
 (3) <i>Oscillatoria</i> (4) All</p> | <p>7. Colonies of BGA is surrounded by :
 (1) Gelatinous sheath
 (2) Hydrocolloids
 (3) Pectin
 (4) Cellulose</p> <p>8. Cyanobacteria or BGA similar to higher plant having in :
 (1) Chlorophyll-a
 (2) All type of chlorophyll
 (3) Cell wall composition
 (4) Food storage</p> <p>9. Heterocyst is :
 (1) Photosynthetic and Nitrogen fixation structure
 (2) Photosynthetic and Non-Nitrogen fixation structure
 (3) Non-Nitrogen fixative structure and Non-Photosynthetic
 (4) Nitrogen fixative and Non-Photosynthetic structure</p> <p>10. Fertility of soil is Increased by :-
 (1) Nitrogen - fixing bacteria
 (2) Denitrifying bacteria
 (3) Plasmalemma
 (4) Cell membrane</p> |
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| <p>11. Harmful activity of Blue green algae is-</p> <ol style="list-style-type: none"> (1) Denitrification (2) Water - bloom (3) Increase alkalinity of soil (4) Decrease fertility of soil <p>12. The cyanobacteria are :</p> <ol style="list-style-type: none"> (1) Unicellular, colonial or filamentous marine or terrestrial algae (2) Available only in the form of colonies (3) Sometimes surrounded by mucilaginous / gelatinous sheath (4) Not associated with algal bloom <p>13. Antibiotic mostly obtained from</p> <ol style="list-style-type: none"> (1) Bacteria (2) Viruses (3) Fungi (4) Angiosperm <p>14. Typhoid and tetanus are caused by :-</p> <ol style="list-style-type: none"> (1) Bacteria (2) Virus (3) Viroids (4) Fungi <p>15. Chemicals produced by one microbe and inhibit growth of another microbes are called</p> <ol style="list-style-type: none"> (1) Antibiotics (2) Aflatoxin (3) Antibodies (4) Phytoalexin | <p>16. Heterocyst is a structure which is associated with</p> <ol style="list-style-type: none"> (1) Reproduction (2) Respiration (3) Nitrogen fixation (4) Locomotion <p>17. In blue green algae photosynthesis occurs at :-</p> <ol style="list-style-type: none"> (1) Chromatophore (2) Chloroplast (3) Leucoplast (4) Chromoplast <p>18. During the rainy season ground surface become slippery due to :-</p> <ol style="list-style-type: none"> (1) Fungi (2) Blue green algae (3) Bryophytes (4) Slime molds <p>19. True sexual reproduction absent in :-</p> <ol style="list-style-type: none"> (1) Yellow green algae (2) Red algae (3) Green algae (4) Blue green algae |
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Mycoplasma

- PPLO is known as a joker of plant kingdom or joker of microbiology.
- PPLO is pleomorphic organism
- They can't be filtered by bacterial proof filter.

Main Points :

1. Mycoplasmas unicellular, smallest prokaryotic organisms. The diameter of cell varies from 0.1mm to 0.3mm.
2. Cell membrane is tri-layered and made up of lipoprotein. Both DNA (ds DNA linear) and RNA are present.
3. Mycoplasmas may be the simplest form of life capable of independent growth, reproduction and metabolism.
4. They are cell wall less prokaryotes.
5. They are resistant to antibiotics as penicillin that act on cell wall.
6. They are sensitive to tetracycline & chloramphenicol that act on metabolic activities.
7. They can survive in absence of Oxygen.

Exercise 1.6

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| <p>1. <i>Mycoplasma</i> is :</p> <p>(1) Facultative anaerobs</p> <p>(2) Obligate anaerobs</p> <p>(3) Facultative aerobic</p> <p>(4) Obligate aerobic</p> <p>2. <i>Mycoplasma</i> causes disease in :-</p> <p>(1) Plant only</p> <p>(2) Animal only</p> <p>(3) Virus</p> <p>(4) Both plant and animal</p> | <p>3. Which of the following is wall less smallest organism?</p> <p>(1) Eubacteria</p> <p>(2) <i>Mycoplasma</i></p> <p>(3) Cyanobacteria</p> <p>(4) Archaeobacteria</p> <p>4. Prokaryotic smallest cell is :-</p> <p>(1) <i>Mycoplasma</i></p> <p>(2) Cyanobacteria</p> <p>(3) Bacteria</p> <p>(4) <i>Bacillus</i></p> |
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Kingdom Protista

All the organism included in Protista are unicellular (acellular) eukaryotes. Members of protista are primarily aquatic. This kingdom forms a link with the others dealing with plants, animals and fungi. There is no well defined boundary of kingdom Protista

Following organism are included in Protista : Dinoflagellates, Chrysophytes, Euglenoids, Slime molds, Protozoans

General characters of Protista :

1. Unicellular, eukaryotic organisms.
2. Mostly aquatic organisms.
3. Cell structure is eukaryotic type having all kinds of membrane bound organelles and 80S cytoplasmic ribosomes and cells may possess cellulosic cell wall.
4. Flagella and cilia have (9+2) pattern of microtubule organization consisting of tubulin protein.
5. Movement by pseudopodia, flagella or cilia where ciliary mode is most efficient.
6. They show highly diverse mode of nutrition.

Reproduction :

1. Vegetative Reproduction :

This is the most common method of reproduction in protists. Vegetative reproduction takes place in favourable condition.

Binary fission : Two daughter cells are formed by the division of one mother cell. After this each daughter cell grows to form a normal organism.

2. Asexual Reproduction :

Spore formation :- Some protists have special structure known as sporangia. Spores are formed in this sporangia. The sporangia bursts after sometime and all the spores become free. These spores form a new cell after germination.

3. Sexual Reproduction :

Sexual reproduction was first of all seen in protists. In sexual reproduction two haploid gametes fuse to form a diploid zygote. This process is known as **syngamy**.

Exercise 1.7

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| <p>1. Single cell eukaryotic organisms are placed in :
 (1) Kingdom fungi
 (2) Kingdom Protista
 (3) Kingdom plantae
 (4) Kingdom Monera</p> <p>2. In all the Five kingdom which kingdom has not well defined boundaries ?
 (1) Kingdom Protista
 (2) Kingdom fungi
 (3) Kingdom plantae
 (4) Kingdom Monera</p> <p>3. Which one of the following kingdom is classified under Five kingdom classification as prokaryotes?
 (1) Monera (2) Fungi
 (3) Plantae (4) Protista</p> | <p>4. Members of Protista primarily are -
 (1) Terrestrial
 (2) Aquatic
 (3) Prokaryotic
 (4) All of the above</p> <p>5. According to five kingdom system the unicellular algae like Dinoflagellates, Diatoms and Euglenoids are included in kingdom :-
 (1) Monera (2) Protista
 (3) Plantae (4) Animalia</p> <p>6. Protists which are diploid reproduce sexually by the process of -
 (1) Zygotic meiosis (2) Cyst formation
 (3) Binary fission (4) gametic meiosis.</p> <p>7. Link between prokaryotes and multicellular eukaryotes :-
 (1) Cyanobacteria (2) Protista
 (3) Fungi (4) Plants</p> |
|---|--|

Division – Chrysophyta

This group includes diatoms and golden algae (desmids). They are found in fresh water as well as in marine environment.

Structure :

- They are found in different shapes such as circular, rectangular, triangular, elongated and boat shaped.
- The cell wall of diatoms is made up of cellulose in which silica particles are embedded in at many places. Due to which the cell wall appears to be made up of silica. This silicated cell wall is called "shell" or "frustule".

Special Note:-

Their cell wall is made up of two halves, which are arranged like soap case. Their cell wall have silica in very large quantity. Due to this reason their cell wall is hard and indestructible. The cell wall does not get destroyed after their death so at the bottom of sea, very huge rocks of dead diatoms are formed which are known as "**diatomite**" or "**diatomaceous earth**" or "**keiselguhr**".

- Diploid nucleus is present in Diatom.
- Diatoms have pigments are **Chlorophyll 'a', Chl 'c', and xanthophyll (fucoxanthin)**. Due to these pigments it appears golden coloured.
- They are chief producers in the ocean.

Special Note:-

- Stored food : Leucosin (Chrysolaminarian) & fats (Oil).**
- Movement :** They are immotile, because flagella are absent in them. They float on the surface of water with the help of low molecular weight **stored fats**.

- (2) Slime moulds are saprophytes found on decaying twigs and leaves engulfing organic material.
- (3) Their body is made up of wall less multinucleated protoplasmic mass. This type of body is known as **plasmodium**. (Plasmodium = wall less coenocyte). Under suitable conditions they form an aggregation which may grow and spread over several feet. During unfavourable conditions plasmodium forms fruiting bodies, which bear spores at their tip.
- (4) They are diploid i.e. every nucleus found in the cell is diploid **E.g. *Physarum*, *Physarella*, *Fuligo*.**

Special Note:-

They lack cell wall in vegetative stage but at the time of reproduction they form cell wall.

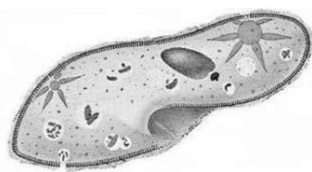
Exercise 1.8

1. Which of the following protist is responsible for red tides ?
 (1) Chrysophytes (2) Dinoflagellates
 (3) Euglenoids (4) Slime moulds
2. Soap box like structure found in :
 (1) *Diatoms* (2) *Gonyaulax*
 (3) Euglenoids (4) Slime moulds
3. Spore dispersal mechanism by air is found in :
 (1) *Diatoms* (2) *Gonyaulax*
 (3) Euglenoids (4) Slime moulds
4. True cell wall in spores of slime moulds is formed under :
 (1) Unfavourable conditions
 (2) Favourable conditions
 (3) Any conditions
 (4) Both (1) and (2)
5. Which of the followings are photosynthetic ?
 (1) Chrysophytes (2) Dinoflagellates
 (3) Euglenoids (4) All of the above
6. Which of the following are mostly marine ?
 (1) *Gonyaulax* (2) Diatoms
 (3) *Euglena* (4) Slime moulds
7. Chief producer in oceans is :
 (1) Diatom (2) Dinoflagellates
 (3) *Euglena* (4) Slime moulds
8. The diatomaceous earth is used to insulate boilers and steam pipes because:-
 (1) The wall of diatoms is deposited with calcium
 (2) The diatomaceous earth is cheap
 (3) It is a good conductor of heat
 (4) The wall of diatoms is made of silica
9. Protist used for the construction of sound proof rooms, is -
 (1) Dinoflagellate (2) Diatoms
 (3) Euglenoids (4) Zooflagellates
10. The diatoms do not easily decay like most the other Algae because:-
 (1) They have water proof cells
 (2) Their walls are mucilagenous
 (3) They have highly siliceous wall
 (4) They are non living
11. "Keiselgurh" a heat resistant material is obtained from :-
 (1) Red Algae (2) Brown Algae
 (3) Diatoms (4) Fungi
12. Dinoflagellates are called fire algae due to which character :-
 (1) They appear like fire due to pigments
 (2) They produce fire due to friction
 (3) They occur on burnt places
 (4) They show bioluminescence
13. Shell of diatoms is made up of :-
 (1) Silica (2) Calcium carbonate
 (3) Keratin (4) Calcium oxalate
14. The dead remains of diatoms are known as :-
 (1) Coenobium (2) Sporangia
 (3) Kieselgurh (4) Sporocarp
15. Decomposer protist are:
 (1) Diatoms (2) Dinoflagellates
 (3) Slime moulds (4) Euglenoid
16. Toxins (Saxitoxins) secreted by some dinoflagellates enter the body of human beings through food chain and result in-
 (1) Madness (2) Paralysis
 (3) Syphilis (4) Plague

- | | |
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| <p>17. Most characteristic feature of diatoms is:-
 (1) Pigments
 (2) Stored food
 (3) Cell wall
 (4) Non oxygenic photosynthesis</p> <p>18. "Naked fungi" (Slime moulds) are naked :-
 (1) When they are reproducing
 (2) When they are not reproducing
 (3) Always naked
 (4) Never naked</p> <p>19. Organism of which kingdom feed like animals and perform photosynthesis like plants :-
 (1) Monera
 (2) Protista
 (3) Mycota
 (4) Animalia</p> | <p>20. Diatoms float in water due to :-
 (1) Stored fats
 (2) Gas Vacuoles
 (3) Flagella
 (4) Air chambers</p> <p>21. Armoured cell wall and biflagellated cells are characteristic of :-
 (1) Diatoms
 (2) Dinoflagellates
 (3) Euglenophyta
 (4) BGA</p> <p>22. "Golden Algae" is the common name of Algae belonging to :
 (1) Diatoms
 (2) Dinoflagellates
 (3) Euglenophyta
 (4) BGA</p> |
|---|---|

Phylum - Protozoa

- They are world wide, Cosmopolitan mostly **Microscopic**, mostly **Aquatic, and free living (*Amoeba*)** or **parasitic (*Plasmodium*)**. These causes serious diseases.
- All protozoans are heterotrophs and live as predators or parasites. **Nutrition** of Protozoans are mainly **holozoic** (*Amoeba*), **Mixotrophic**. (*Euglena*), **Parasitic (*Plasmodium*)**.
- Reproduction :**
 - Asexual** - Binary fission, Multiple fission
 - Sexual** - Syngamy, Conjugation
- They are believed to be primitive relatives of animals. There are four major groups of protozoans.
 - Amoeboid protozoans:** These organisms live in fresh water, sea water or moist soil. They move and capture their prey by putting out pseudopodia (false feet) as in *Amoeba*. Marine forms have silica shells on their surface. Some of them such as *Entamoeba* are parasites.
 - Flagellated protozoans:** The members of this group are either free-living or parasitic. They have flagella. The parasitic forms cause diseases such as sleeping sickness. Example: *Trypanosoma*.
 - Ciliated protozoans:** These are aquatic, actively moving organisms because of the presence of thousands of cilia. They have a cavity (gullet) that opens to the outside of the cell surface. The coordinated movement of rows of cilia causes the water laden with food to be steered into the gullet. Example: *Paramecium*.



- Sporozoans:** This includes diverse organisms that have an infectious spore-like stage in their life cycle. The most notorious is *Plasmodium* (malarial parasite) which causes malaria, a disease which has a staggering effect on human population.

Exercise 1.9

1. Fungi are grouped into various classes based on ?
 (1) Morphology of mycelium
 (2) Mode of spore formation
 (3) Fruiting bodies
 (4) All of these
2. Mostly fungi are
 (1) Heterotrophic (2) Photosynthetic
 (3) Chemoautotrophic (4) Predator
3. Body of fungi is called :
 (1) Mycellium (2) Hypha
 (3) Group of hyphae (4) Haustoria
4. Which of the following is unicellular fungi ?
 (1) Yeast (2) *Puccinia*
 (3) *Penicillium* (4) *Rhizopus*
5. Select the odd one w.r.t. Ascomycetes
 (1) *Aspergillus* (2) *Neurospora*
 (3) *Claviceps* (4) *Colletotrichum*
6. Cell wall of fungi made up of
 (1) Chitin only
 (2) Chitin and polysaccharides
 (3) Polysaccharides only
 (4) Starch
7. Which of the following is sexual spore in fungi ?
 (1) Oospores, ascospores, basidiospores
 (2) Oospores, ascospores, zoospores
 (3) Ascospores, zoospore, basidiospores
 (4) Sporangiospores, ascospores, basidiospores
8. Fungi shows great diversity in :
 (1) Morphology only
 (2) Habitat only
 (3) Reproduction
 (4) Morphology and habitat
9. Which of the following is parasitic fungi ?
 (1) *Rhizopus* (2) *Albugo*
 (3) Yeast (4) Morels
10. Which of the following classes of fungi have dikaryon condition ?
 (1) Ascomycetes and basidiomycetes
 (2) Ascomycetes and deuteromycetes
 (3) Basidiomycetes and phycomycetes
 (4) All of these
11. Organism group which shows Heterotrophic mode of nutrition is :
 (1) Algae (2) Fungi
 (3) Bryophytes (4) Pteridophytes
12. In fungi lump of hyphae is referred to as :-
 (1) Parenchyma (2) Haustorium
 (3) Mycelium (4) Carpogonium
13. Stored food material of fungi :-
 (1) Cellulose
 (2) Starch
 (3) Glycogen and starch
 (4) Glycogen and oil
14. Heterotrophic and absorptive mode of nutrition is found in :-
 (1) Algae (2) Fungi
 (3) Bryophytes (4) Euglenoid
15. The fungi are :-
 (1) Autotrophic (2) Holotrophic
 (3) Chemotrophic (4) Heterotrophic
16. Non-septate mycelium occurs in:-
 (1) Phycomycetes (2) Ascomycetes
 (3) Basidiomycetes (4) Deuteromycetes
17. All fungi are :-
 (1) With chlorophyll
 (2) Without chlorophyll
 (3) With carotene
 (4) Wall less
18. In fungi, vegetative reproduction takes place by :
 (1) Fragmentation (2) Conidia
 (3) Zygosporangium (4) Oospore
19. In ascomycetes and basidiomycetes, the dikaryotic cell is :
 (1) Haploid (2) Diploid
 (3) Triploid (4) Polyploid
20. Coenocytic mycelium is found in :-
 (1) *Rhizopus* (2) *Mucor*
 (3) *Penicillium* (4) Both (1) and (2)

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| <p>21. Fungi are ecologically important because -</p> <ol style="list-style-type: none"> (1) They yield antibiotics (2) They are used in genetic studies (3) They function as decomposers (4) All of the above <p>22. Which of the following characters indicate similarity between fungi and animals ?</p> <ol style="list-style-type: none"> (1) Heterotrophic nutrition (2) Type of stored food (3) Presence of chitin (4) All of the above | <p>23. The fungus without mycelium is -</p> <ol style="list-style-type: none"> (1) Phytophthora (2) Rhizopus (3) Saccharomyces (4) Microsporum <p>24. Nucleated spore bearing non chlorophyllous organism belong to:</p> <ol style="list-style-type: none"> (1) Monera (2) Thallophyta (3) Fungi (4) Bryophyta |
|--|--|

Phycomycetes:

- All the fungi included in this class are called as lower fungi. Members of phycomycetes are found in aquatic habitat (members of this class known as algal fungi) and on decaying wood in moist and damp place or as obligate parasites on plants.

Mycelium :-

- The fungal filament (mycelium) of all the fungus included in this class are coenocytic and aseptate and branched.
- Phycomycetes further classified into oomycetes and zygomycetes

Note:

The mycelium of oomycetes and zygomycetes classes is same in structure i.e. coenocytic & aseptate. But they are dissimilar on the basis of sexual reproduction.

(a) Oomycetes :

Asexual reproduction : By the formation of sporangiospores (zoospores) & conidia.

Sexual reproduction : Oospore (thick walled)

There is no time interval between plasmogamy and karyogamy, so dikaryophase is not formed.

E.g. (1) *Phytophthora infestans* : Causes "late blight of potato".

(2) *Sclerospora graminicola* : Causes "Green ear disease" of Bajra.

(3) *Albugo candida* : Parasite on mustard.

(b) Zygomycetes :-

Asexual reproduction – By the formation of sporangiospores (aplanospores).

Sexual reproduction – By Zygosporangium (thick walled).

Types of sexual reproduction

- Isogamous – All the members

E.g. (1) *Rhizopus*

(2) *Mucor*

3. Which of the following is coprophilous fungi ?
(1) Club fungi (2) Sac fungi
(3) Algal fungi (4) Fungi imperfecti
4. Which of the following is **not** applicable for fungi ?
(1) They are eukaryotic
(2) They are heterotrophic
(3) They are both unicellular and multicellular
(4) They possess a purely cellulosic cell wall
5. Storage grains come to have aflatoxin due to growth of :
(1) *Aspergillus flavus* (2) Yeast
(3) Bacterium (4) Virus
6. Which one of the following has photosynthetic pigment similar to higher plants :
(1) Diatoms (2) Dinoflagellates
(3) Euglenoids (4) Green algae
7. The basidiomycetes includes :-
(1) Rusts (2) Smuts
(3) Mushrooms (4) All of the above
8. For bread and beer making, commonly :
(1) Unicellular algae are used
(2) Multicellular algae are used
(3) Unicellular fungi are used
(4) Multicellular fungi are used
9. The fungi which grow on dung is called :-
(1) Epixylic (2) Coprophilous
(3) Keratinophilic (4) Xylophilous
10. Which of the following causes wheat rust disease:-
(1) A red Algae (2) A green Algae
(3) A fungi (4) Mycoplasma
11. Deuteromycetes are called 'Imperfect fungi' as :-
(1) They have no cell wall
(2) No mycelium
(3) No sexual reproduction
(4) No asexual reproduction
12. Sexual reproduction absent in :-
(1) Phycomycetes (2) Deuteromycetes
(3) Zygomycetes (4) Basidiomycetes
13. Which of the following is not a member of class phycomycetes?
(1) *Albugo* (2) *Mucor*
(3) *Rhizopus* (4) *Neurospora*
14. Aspergillosis is caused by :-
(1) Virus (2) Bacteria
(3) Fungi (4) Mycoplasma
15. One of the common fungal disease of human is :-
(1) Cholera (2) Plague
(3) Typhoid (4) Ring worm
16. *Neurospora*, which is popularly known as Drosophila of plant kingdom, belongs to :-
(1) Phycomycetes (2) Ascomycetes
(3) Basidiomycetes (4) Deuteromycetes
17. Which of the following is a form class ?
(1) Deuteromycetes (2) Basidiomycetes
(3) Rhodophyceae (4) Euglenophyceae
18. The sac fungi belongs to :-
(1) Ascomycetes (2) Basidiomycetes
(3) Phycomycetes (4) Deuteromycetes
19. Which of the following secretes toxins during storage conditions of crop plants?
(1) *Aspergillus* (2) *Penicillium*
(3) *Fusarium* (4) *Colletotrichum*
20. The chief characteristic of class Ascomycetes is:-
(1) Formation of spores
(2) Hyphae
(3) Formation of aco-spores
(4) Formation of zoospores
21. Yeast grows more quickly in -
(1) Salt water
(2) Sugar solution
(3) Double distilled water
(4) Marine water
22. A. Flemming isolated penicillin from -
(1) *P. chrysogenum* (2) *P. notatum*
(3) *Aspergillus Flovus* (4) *A. niger*
23. Alexander Flemming discovered penicillin in 1928 while working with-
(1) *Streptomyces*
(2) Bacteria (*Staphylococcus*)
(3) *Penicillium notatum*
(4) *P. chrysogenum*
24. Pseudomycelium occurs in -
(1) Mushroom (2) *Mucor*
(3) Bread mold (4) Yeast

Mycorrhiza

- Mycorrhiza is the symbiotic association between roots of higher plants and fungus. Fungus obtain food and shelter from root and fungi helps to root in absorption of water and mineral and also protect plant from soil borne pathogen.
- It is useful for **phosphate absorption**.

Exercise 1.11

- "Potato spindle tuber disease" refers to :
(1) Viroids (2) Prions
(3) Virus (4) Bacteria
- Viroids differ from virus in :-
(1) Protein coat (2) DNA
(3) RNA (4) All
- BSE disease is caused by
(1) Virus (2) Viroids
(3) Prions (4) Lichens
- In plants symptoms of viral disease are
(1) Mosaic formation
(2) Leaf rolling and curling
(3) Yellowing and Vein clearing
(4) All
- 'Contagium vivium fluidum' belongs to
(1) Virus (2) Viroids
(3) Prions (4) Lichens
- Which of the following organism can not grow in polluted area ?
(1) Virus (2) Fungi
(3) Bacteria (4) Lichens
- Which of the following are not considered in five kingdom system of classification?
(1) Lichen (2) Virus
(3) Viroid (4) All of the above
- Select the true statement :
(1) Viruses are obligate parasites
(2) In lichens, algal component is called phycobiont
(3) In lichens, fungal component is called mycobiont
(4) All of the above
- Viroid was discovered by:
(1) Haeckel (2) Pasteur
(3) T.O. Diener (4) Theophrastus
- The protein coat of viruses are called _____ which is made of small subunits called _____.
(1) Capsomere, Capsid
(2) Pepleomere, Capsid
(3) Capsid, Pepleomere
(4) Capsid, Capsomere
- The function of mycobiont in a lichen is to :
(1) Prepare food for algae
(2) Provide shelter and absorb nutrients and water for algae
(3) Digest algae
(4) All are incorrect
- Which of the following is non-cellular organism?
(1) Bacteria (2) Fungi
(3) Algae (4) Virus
- Enzymes are **not** found in
(1) Fungi (2) Algae
(3) Virus (4) Cyanobacteria
- Lichens show:
(1) Mutually obligatory relationship
(2) Mutually non-obligatory relationship
(3) Mutually relationship between algae and higher plants
(4) Relationship between gymnosperm and BGA
- Phycobionts and mycobionts are respectively :
(1) Autotrophic, Heterotrophic
(2) Heterotrophic, Autotrophic
(3) Autotrophic, Autotrophic
(4) Heterotrophic, Heterotrophic
- Lichens are pollution indicator for :
(1) SO₂ (2) CO₂
(3) NO₂ (4) SiO₂
- VAM is associated to :
(1) Ectomycorrhiza (2) Endomycorrhiza
(3) Lichens (4) Both (1) and (2)

Exercise 2

1. Archaeobacteria differ from other bacteria in having a :
 - (1) Different cell wall structure
 - (2) Different cellular organization
 - (3) Parasitic nature
 - (4) All of these
2. Which character indicates that cyanobacteria are similar to higher green plants ?
 - (1) Type of cell wall
 - (2) Nitrogen fixation ability
 - (3) Presence of chlorophyll 'a'
 - (4) Presence of gelatinous sheath
3. Which one of the following prokaryotes lacks cell wall:
 - (1) Virus
 - (2) Cyanobacteria
 - (3) Mycoplasma
 - (4) Protozoa
4. Which organisms have silica in their cell wall ?
 - (1) Dinoflagellates
 - (2) Slime mould
 - (3) Diatoms
 - (4) Euglenoids
5. Which organism is responsible for the red surface of the sea ?
 - (1) Euglena
 - (2) Gonyaulax
 - (3) Amoeba
 - (4) Paramecium
6. Read the following statements:
 - (i) The body organization of fungi is multicellular/loose tissue type
 - (ii) Mode of nutrition in monerans is autotrophic or heterotrophic
 - (iii) Nuclear membrane may be present in E. coli
 - (iv) SCell wall is cellulosic in animals cells
 - (v) Non-cellulosic cell wall is present in fungi
 How many of the above statements are **correct** ?
 - (1) Three
 - (2) Two
 - (3) Four
 - (4) Five
7. Read the following statements :
 - (i) The bacterial structure is very simple but they are complex in behaviour
 - (ii) Compared to many other organisms, bacteria as a group show the most extensive metabolic diversity
 - (iii) Some of the bacteria are autotrophic
 - (iv) Most of the bacteria are heterotrophic
 - (v) Bacteria possess cell wall How many of the above statements are **not incorrect**?
 - (1) Four
 - (2) Five
 - (3) Two
 - (4) Zero
8. Read the following statements :-
 - (i) Majority of the slime moulds are fresh water organisms
 - (ii) Dinoflagellates are mostly fresh water organisms
 - (iii) Slime moulds form plasmodium
 - (iv) Protozoans may be autotrophs
 - (v) The spores of slime moulds are dispersed by water currents
 Which of the above statements are **correct**?
 - (1) Only (iii)
 - (2) All five
 - (3) (ii), (ii) and (v)
 - (4) (i), (ii), (iv) and (v)
9. Select the correct statement about slime moulds :
 - (1) Saprophytic, plasmodium with cell wall, spores dispersed by air currents
 - (2) Saprophytic, plasmodium without cell wall, spores with true cell wall and dispersed by air currents
 - (3) Parasitic, plasmodium without cell wall, spores dispersed by air current
 - (4) Parasitic, plasmodium without cell wall, spores with true cell wall and dispersed by water current
10. The genetic material of prokaryotic cell :
 - (1) Non-histonic double-stranded DNA
 - (2) Histonic double stranded DNA
 - (3) Histone & DNA both are absent
 - (4) Histone without DNA

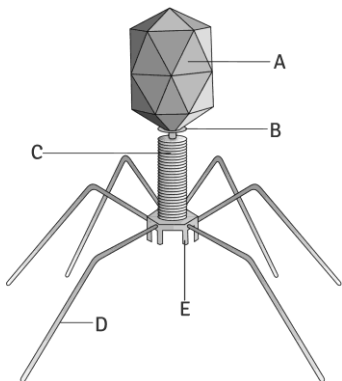
11. In bacteria, the plasmid is:
 (1) Extra chromosomal material
 (2) Main DNA
 (3) Non functional DNA
 (4) Repetative gene
12. In which kingdom would you classify the archaea and nitrogen-fixing organisms, if the five-kingdom system of classification is used :-
 (1) Plantae (2) Fungi
 (3) Protista (4) Monera
13. Carbohydrates are the most abundant biomolecules on earth. are produced by?
 (1) Some bacteria, algae and green plant cells
 (2) All bacteria, fungi and algae
 (3) Fungi, algae and green plant cells
 (4) Viruses, fungi and bacteria
14. Match the columns-I and II, and choose the **correct** combination from the options given.

	Column-I (Kingdom)		Column-II (Mode of Nutrition)
(a)	Monera	(i)	Autotrophic
(b)	Protista	(ii)	Heterotrophic
(c)	Fungi	(iii)	Autotrophic and heterotrophic both
(d)	Plantae		
(e)	Animalia		

- (1) a-iii, b-iii, c-ii, d-i, e-ii
 (2) a-ii, b-i, c-ii, d-iii, e-ii
 (3) a-iii, b-ii, c-iii, d-i, e-iii
 (4) a-ii, b-iii, c-ii, d-iii, e-ii
15. Some characters are given below. Which of them is **true** for the cyanobacteria ?
 (A) Oxygenic photosynthesis
 (B) Non-oxygenic chemotrophism
 (C) Oxygenic chemotrophism
 (D) Ability of N_2 fixation
 (1) A and B (2) A and C
 (3) A and D (4) B and D

16. Select the **incorrect** statement about fungi?
 (1) The cell wall is composed of chitin and polysaccharides
 (2) These are heterotrophic organisms
 (3) They are autotrophic organisms
 (4) They are absorptive in nature
17. The mycelium is aseptate and Coenocytic in :
 (1) Phycomycetes
 (2) Basidiomycetes
 (3) Deuteromycetes
 (4) Ascomycetes
18. Septate and branched mycelium are found in
 (1) Basidiomycetes
 (2) Deuteromycetes
 (3) Ascomycetes
 (4) All of the above
19. Select the **correct** matched pair/pairs :
 (1) Claviceps – Ascomycetes
 (2) Morels – Ascomycetes
 (3) Agaricus – Basidiomycetes
 (4) All the above
20. Asexual reproduction takes place by conidia formation in :-
 (1) Aspergillus (2) Penicillium
 (3) Colletotrichum (4) All of the above
21. Read the following statements w.r.t. fungi :
 (i) Their body consists of long, slender thread like structures called hyphae
 (ii) The network of hyphae is called mycelium
 (iii) Hyphae may be continuous tubes filled with multinucleated cytoplasm, these hyphae are coenocytic
 (iv) In many cases hyphae have septate or cross walls
 (v) The cell wall of fungi are composed of chitin and polysaccharides
 How many of the above statements are **correct** ?
 (1) One (2) Three
 (3) Four (4) Five

22. Identify the virus and name the structure marked A, B, C, D and E :



- (1) Bacteriophage, A= Head, B= Sheath, C= Collar, D= Tail fibre, E= Tail pin
 (2) Bacteriophage, A= Head, B= Collar, C= Base plate, D= Sheath, E= Base Tail fibre
 (3) Bacteriophage, A= Head, B= Sheath, C= Tail fibre, D= Collar, E= Tail pin
 (4) Bacteriophage, A= Head, B= Collar, C= Sheath, D= Tail fibre, E= Tail pin
23. Read the names of different fungi given below :
 Mucor, Albugo, Penicillium, Ustilago, Colletotrichum, Puccinia, Agaricus, Aspergillus and Rhizopus
 How many of the following belong to the class phycomycetes ?
 (1) Three (2) Two
 (3) Four (4) Seven
24. In oogamy, fertilization involves :-
 (1) A large motile female gamete and a small non motile male gamete
 (2) A small nonmotile female gamete and a large motile male gamete
 (3) A large nonmotile female gamete and a small motile male gamete
 (4) A large non motile female gamete and a small nonmotile male gamete
25. Find the incorrect statement about fungi :
 (1) They show a great diversity in morphology and habitats
 (2) Fungi are cosmopolitan and occur in air, water soil and as parasites also
 (3) They prefer to grow in cold and dry places
 (4) With the exception of yeasts, fungi are filamentous

26. Select the correct statements about fungi :
 (A) Most of the fungi are saprophytes
 (B) Fungi can also survive as symbionts
 (C) Sexual reproduction in fungi involves three steps
 (D) Reduction division in fungi occurs in fruiting bodies which forms haploid spores
 (1) A, B are correct
 (2) C, D are correct
 (3) A, B & C are correct
 (4) A, B, C & D are correct
27. Choose the incorrect statement regarding the decomposer :
 (1) They may be prokaryotes or eukaryotes.
 (2) They may be unicellular or multicellular.
 (3) They convert inorganic into organic compounds
 (4) They play a great role in ecology
28. With respect to fungal sexual cycle, what is the **correct** sequence :
 (1) Karyogamy, Plasmogamy & Meiosis
 (2) Meiosis, Plasmogamy & Karyogamy
 (3) Plasmogamy, Karyogamy & Meiosis
 (4) Meiosis, Karyogamy & Plasmogamy
29. Prokaryotic cell is characterized by :-
 (1) Presence of nucleus, mitochondria and plastids
 (2) Absence of cell wall, DNA fibrils and plastids
 (3) Presence of spindle fibres, DNA fibrils and golgi bodies
 (4) Absence of endoplasmic reticulum, golgi bodies and spindle fibres.
30. Dinoflagellates have two flagella :
 (1) Both lying longitudinally between the wall plates.
 (2) One lying longitudinally and the other transversely in a furrow between the wall plates.
 (3) Both lying transversely between the wall plates
 (4) Do not help in their movement

31. Euglenoids :

- (1) Are photosynthetic in the presence of sunlight but when deprived of sunlight they behave like heterotrophs by predating other smaller organisms
- (2) Possess carbohydrate rich pellicle which makes their body flexible
- (3) They possess heterokont flagella i.e., one short and the other long
- (4) More than one option is correct

32. The thalloid body of a slime mould (Myxomycetes) is known as :-

- (1) Fruiting body
- (2) Mycelium
- (3) Protonema
- (4) Plasmodium

33. Match the entities in column-I with their character in column-II:

	Column-I		Column-II
(a)	Protozoans	(i)	Diatoms and golden algae
(b)	Slime Moulds	(ii)	Gonyaulax
(c)	Euglenoids	(iii)	Trypanosoma
(d)	Chrysophytes	(iv)	Saprophytic protists
(e)	Dinoflagellates	(v)	Euglena

- (1) a-iii, b-iv, c-v, d-ii, e-i
- (2) a-iii, b-iv, c-v, d-i, e-ii
- (3) a-iii, b-v, c-iv, d-ii, e-i
- (4) a-iv, b-v, c-ii, d-iii, e-i

34. Match the columns-I and II, and choose the correct combination from the options given.

	Column-I		Column-II
(a)	Phycomycetes	(i)	Sac fungi
(b)	Ascomycetes	(ii)	Imperfect fungi
(c)	Basidiomycetes	(iii)	Algal fungi
(d)	Deuteromycetes	(iv)	Club fungi/ bracket fungi

- (1) a-i, b-iii, c-iv, d-ii (2) a-iii, b-i, c-iv, d-ii
- (3) a-i, b-iii, c-ii, d-iv (4) a-iii, b-i, c-ii, d-iv

35. Match the columns-I and II, and choose the correct combination from the options given.

	Column-I		Column-II
(a)	Morels	(i)	Deuteromycetes
(b)	Smut	(ii)	Ascomycetes
(c)	Bread mould	(iii)	Basidiomycetes
(d)	Imperfect fungi	(iv)	Phycomycetes

- (1) a-iii, b-iv, c-i, d-ii (2) a-ii, b-iii, c-iv, d-i
- (3) a-iv, b-i, c-ii, d-iii (4) a-iii, b-iv, c-ii, d-i

36. Match the columns-I and II, and choose the correct combination from the options given.

	Column-I		Column-II
(a)	Eubacteria	(i)	Trichoderma
(b)	Dinoflagellates	(ii)	Albugo
(c)	Phycomycetes	(iii)	Gonyaulax
(d)	Deuteromycetes	(iv)	Anabaena

- (1) a-i, b-ii, c-iii, d-iv (2) a-ii, b-iii, c-iv, d-i
- (3) a-iv, b-iii, c-ii, d-i (4) a-iii, b-iv, c-i, d-ii

37. Match the columns-I and II, and choose the correct combination from the options given.

	Column-I (Organism)		Column-II (Common name)
(a)	Puccinia	(i)	Yeast
(b)	Ustilago	(ii)	Mushroom
(c)	Agaricus	(iii)	Smut fungus
(d)	Saccharomyces	(iv)	Rust fungus

- (1) a-i, b-ii, c-iii, d-iv (2) a-ii, b-iii, c-iv, d-i
- (3) a-iii, b-iv, c-i, d-ii (4) a-iv, b-iii, c-ii, d-i

38. Match the columns-I and II, and choose the correct combination from the options given.

	Column-I (Scientists)		Column-II (Contribution)
(a)	Ivanowsky	(i)	Contagium vivum fluidum
(b)	Pasteur	(ii)	Crystallisation of viruses
(c)	Beijerinck	(iii)	Discovery of viruses
(d)	Stanley	(iv)	Term virus

- (1) a-iii, b-iv, c-ii, d-i (2) a-iv, b-iii, c-ii, d-i
- (3) a-iii, b-iv, c-i, d-ii (4) a-iv, b-iii, c-i, d-ii

39. Match the columns I and II, and choose the **correct** combination from the option given.

	Column-I (Types of)		Column-II (Production)
(a)	Zoospores	(i)	Exogenous
(b)	Aplanospores	(ii)	Endogenous
(c)	Conidia		
(d)	Ascospores		
(e)	Basidiospores		

- (1) a-i, b-ii, c-i, d-ii, e-ii
 (2) a-ii, b-i, c-ii, d-i, e-ii
 (3) a-ii, b-ii, c-i, d-ii, e-i
 (4) a-i, b-ii, c-i, d-ii, e-i

40. Match the columns-I and II, and choose the **correct** combination from the options given.

	Column-I (Group Protista)		Column-II (Example)
(a)	Chrysophytes	(i)	Paramoecium
(b)	Dinoflagellates	(ii)	Euglena
(c)	Euglenoids	(iii)	Gonyaulax
(d)	Protozoans	(iv)	Diatoms

- (1) a-i, b-iii, c-ii, d-iv
 (2) a-ii, b-iv, c-iii, d-i
 (3) a-iv, b-ii, c-iii, d-i
 (4) a-iv, b-iii, c-ii, d-i

Exercise 3

Matrix—Match Type Questions:

1. Match the process in Column I with its term in Column II.

	Column I (Systems of classification)		Column II (Given by)
(A)	Two kingdom system of classification	(p)	RH Whittaker
(B)	Five kingdom system of classification	(q)	Carl Woese
(C)	Six kingdom system of classification	(r)	Carolus Linnaeus

- (1) A → (q); B → (p); C → (r)
 (2) A → (p); B → (q); C → (s)
 (3) A → (s); B → (r); C → (p)
 (4) A → (r); B → (p); C → (q)

2. Match the process in Column I with its term in Column II.

	Column I (Names)		Column II (Shape)
(A)	Coccus	(p)	Rod-shaped
(B)	Bacillus	(q)	Spherical
(C)	Vibrio	(r)	Spiral-shaped
(D)	Spirillum	(s)	Comma-shaped

- (1) A → (r); B → (q); C → (p); D → (s)
 (2) A → (s); B → (r); C → (q); D → (p)
 (3) A → (q); B → (p); C → (s); D → (r)
 (4) A → (p); B → (s); C → (r); D → (q)

3. Match the organisms in Column I with habitats in Column II and choose the correct option from the codes given below.

	Column I		Column II
(A)	Halophiles	(p)	Hot springs
(B)	Thermoacidophiles	(q)	Aquatic environment
(C)	Methanogens	(r)	Guts of ruminants
(D)	Cyanobacteria	(s)	Salty area

- (1) A → (s); B → (p); C → (r); D → (q)
 (2) A → (p); B → (q); C → (r); D → (s)
 (3) A → (r); B → (s); C → (q); D → (p)
 (4) A → (q); B → (s); C → (r); D → (p)

4. Match the organisms in Column I with habitats in Column II and choose the correct option from the codes given below.

	Column I (Features)		Column II (Protista)
(A)	Chief producer in the oceans	(p)	Diatoms
(B)	Red tide	(q)	Dinoflagellates
(C)	Connecting link between plants and animals	(r)	Euglenoids
(D)	Fungus animals	(s)	Slime Moulds

- (1) A → (q); B → (r); C → (s); D → (p)
 (2) A → (p); B → (q); C → (r); D → (s)
 (3) A → (r); B → (s); C → (p); D → (q)
 (4) A → (s); B → (p); C → (r); D → (r)

5. Match the process in Column I with its term in Column II.

	Column I (Types)		Column II (Examples)
(A)	Amoeboid protozoans	(p)	<i>Plasmodium</i>
(B)	Flagellated protozoans	(q)	<i>Paramecium</i>
(C)	Ciliated protozoans	(r)	<i>Trypanosoma</i>
(D)	Sporozoans	(s)	<i>Entamoeba histolytica</i>

- (1) A → (p); B → (q); C → (r); D → (s)
 (2) A → (s); B → (r); C → (q); D → (p)
 (3) A → (r); B → (q); C → (p); D → (s)
 (4) A → (q); B → (p); C → (s); D → (r)

6. Match the process in Column I with its term in Column II.

	Column I (Features)		Column II (Related to)
(A)	Parasitic fungi on mustard	(p)	<i>Neurospora</i>
(B)	Rust and smut disease	(q)	<i>Puccinia</i> and <i>Ustilago</i>
(C)	Used in genetic work	(r)	Morels and truffles
(D)	Edible delicacies	(s)	<i>Albugo</i>
(E)	Bread mould	(t)	<i>Rhizopus</i>

(1) A → (r); B → (t); C → (s); D → (q); E → (p)

(2) A → (p); B → (r); C → (t); D → (s); E → (r)

(3) A → (r); B → (p); C → (r); D → (t); E → (s)

(4) A → (s); B → (q); C → (p); D → (r); E → (t)

7. Match the following columns and choose the correct option from the codes given below

	Column I		Column II
(A)	Saprophyte	(p)	Symbiotic association of fungi with plant roots
(B)	Parasite	(q)	Decomposition of dead organic materials
(C)	Lichens	(r)	Living on living plants or animals
(D)	Mycorrhiza	(s)	Symbiotic association of algae and fungi

(1) A → (r); B → (q); C → (p); D → (s)

(2) A → (q); B → (p); C → (r); D → (s)

(3) A → (q); B → (r); C → (s); D → (p)

(4) A → (p); B → (q); C → (r); D → (s)

8. Match the following columns and choose the correct option from the codes given below :

	Column I (Scientists)		Column II (Related to)
(A)	DJ Ivanowsky (1892)	(p)	Viroids
(B)	MW Beijerinck (1898)	(q)	First crystallised TMV
(C)	WM Stanley (1935)	(r)	<i>Contagium vivum fluidum</i>
(D)	TO Diener (1971)	(s)	Mosaic disease of tobacco

(1) A → (p); B → (s); C → (q); D → (r)

(2) A → (q); B → (p); C → (s); D → (r)

(3) A → (s); B → (r); C → (q); D → (p)

(4) A → (r); B → (q); C → (p); D → (s)

9. Match the following columns and choose the correct option from the codes given below

	Column I (Kingdoms)		Column II (Classes)
(A)	Plantae	(p)	Archaeobacteria
(B)	Fungi	(q)	Euglenoids
(C)	Protista	(r)	Phycomycetes
(D)	Monera	(s)	Algae

(1) A → (s); B → (r); C → (q); D → (p)

(2) A → (p); B → (q); C → (r); D → (s)

(3) A → (r); B → (s); C → (q); D → (p)

(4) A → (s); B → (q); C → (r); D → (p)

Assertion – Reason Type Questions:

10. **Assertion:** The two kingdom classification, used for a long time, was found to be inadequate.

Reason: Two kingdom system of classification did not distinguish between the eukaryotes and prokaryotes, unicellular and multicellular organisms and green algae and fungi.

- (1) If both A and R are true and R is the correct explanation of A
 (2) If both A and R are true, but R is not the correct explanation of A
 (3) If A is true, but R is false
 (4) If A is false, but R is true

- 11. Assertion:** Five kingdom system of classification did not differentiated between the heterotrophic group, fungi and the autotrophic green plants. Though they showed a characteristic difference in their cell wall composition.

Reason: Fungal cell wall contains chitin, while green plants have a cellulosic cell wall.

- (1) If both A and R are true and R is the correct explanation of A
- (2) If both A and R are true, but R is not the correct explanation of A
- (3) If A is true, but R is false
- (4) If A is false, but R is true

- 12. Assertion:** Euglena is a plant due to the presence of chlorophyll.

Reason: Euglena cannot be classified on the basis of two kingdom system.

- (1) If both A and R are true and R is the correct explanation of A
- (2) If both A and R are true, but R is not the correct explanation of A
- (3) If A is true, but R is false
- (4) If A is false, but R is true

- 13. Assertion:** Fungi are wide spread in distribution and they can even live on or inside other plants and animals.

Reason: Fungi are able to grow anywhere on land, water or on other organisms because they have variety of pigments including chlorophyll, carotenoids, fucoxanthin and phycoerythrin.

- (1) If both A and R are true and R is the correct explanation of A
- (2) If both A and R are true, but R is not the correct explanation of A
- (3) If A is true, but R is false
- (4) If A is false, but R is true

- 14. Assertion:** In fungi, sexual apparatus decreases in complexity from lower to higher forms.

Reason: In algae, sexual apparatus increases in complexity from simple to higher forms.

- (1) If both A and R are true and R is the correct explanation of A
- (2) If both A and R are true, but R is not the correct explanation of A
- (3) If A is true, but R is false
- (4) If A is false, but R is true

- 15. Assertion:** Viruses cause diseases and replicate when they are in the host cell.

Reason: Viruses do not replicate outside the host, but they survive in environment.

- (1) If both A and R are true and R is the correct explanation of A
- (2) If both A and R are true, but R is not the correct explanation of A
- (3) If A is true, but R is false
- (4) If A is false, but R is true

- 16. Assertion:** Slime moulds are called as fungus like animals.

Reason: These do not have cell wall.

- (1) If both A and R are true and R is the correct explanation of A
- (2) If both A and R are true, but R is not the correct explanation of A
- (3) If A is true, but R is false
- (4) If A is false, but R is true

- 17. Assertion:** Kingdom-Plantae includes all eukaryotic, chlorophyll containing organisms.

Reason: Few of its members are partially heterotrophic.

- (1) If both A and R are true and R is the correct explanation of A
- (2) If both A and R are true, but R is not the correct explanation of A
- (3) If A is true, but R is false
- (4) If A is false, but R is true

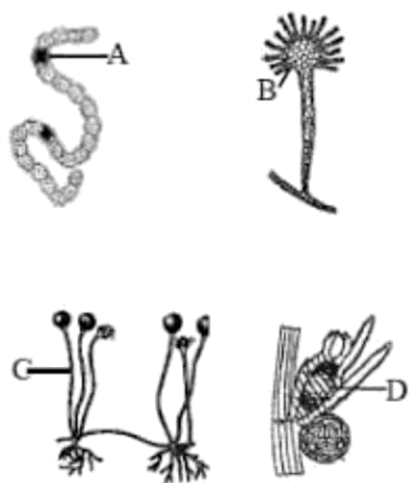
- 18. Assertion:** Three domains system classifies organisms based on cellular characteristics.

Reason: The three domains are, i.e. archaea, bacteria and eukarya

- (1) If both A and R are true and R is the correct explanation of A
- (2) If both A and R are true, but R is not the correct explanation of A
- (3) If A is true, but R is false
- (4) If A is false, but R is true

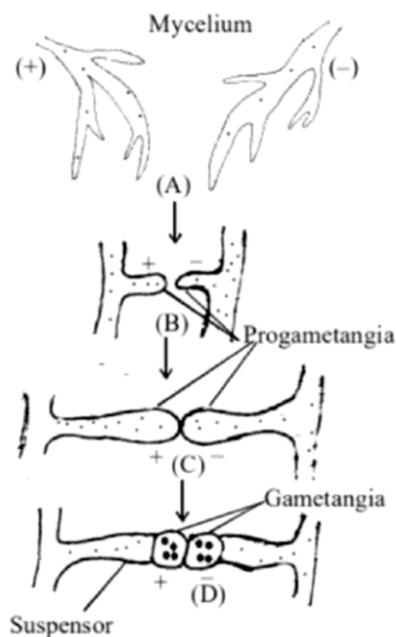
Objective Type Questions:

- 19.** Examine the figures (A-D) given below and select the right option out of 1-4, in which all the four structures A, B, C and D are identified correctly



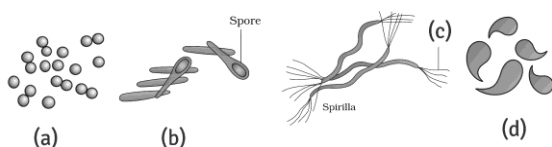
- (1) A – Heterocyst, B - Sterigmata, C - Sporangiphore, D - Oogonium
- (2) A - Akinete, B - Conidia, C - Stolon, D - Antheridium
- (3) A - Heterocyst, B - Conidiophore, C - Stolon, D - Antheridium
- (4) A - Akinete, B - Sterigmata, C - Rhizoidal hyphae, D – Oogonium

- 20.** The figures represent stages in the sexual reproduction in Rhizopus. If gametangia fail to fuse, then what is the fate of cells ?



- (1) They will develop into a zoospore
 - (2) They will develop into a prothallus
 - (3) They will develop into a parthenospore
 - (4) They will develop into a zygospore
- 21.** Study the following statements :
- (A) Bacterial structure is very simple but very complex in behaviour
 - (B) Bacteria show the most extensive metabolic diversity
 - (C) The vast majority of bacteria are autotrophs and a few are heterotrophs
 - (D) Bacteria are not the most abundant microbes in comparison to fungi
- From these, how many statements are incorrect
- (1) One
 - (2) Two
 - (3) Three
 - (4) four

22. Label the a, b, c & d in diagram by correct identification of them :



- (1) (a) - Bacilli bacteria, (b) - Vibrio bacteria, (c) - Flagellum in Bacteria, (d) - Cocci bacteria
 (2) (a) - Cocci bacteria, (b) - Bacilli bacteria, (c) - Flagella in Bacteria, (d) - Comma Bacteria
 (3) (a) - Cocci bacteria, (b) - Heterocyst, (c) - Vibrio bacteria, (d) - Bacilli Bacteria
 (4) (a) - Bacilli bacteria, (b) - Pleomorphic bacteria, (c) - Vibrio bacteria, (d) - Cocci Bacteria

23. What will be the correct/suitable characters in place of A, B, C, D in respective kingdoms in above table ?

S.N.	Character	Monera	Protista	Fungi	Plantae	Animalia
(i)	Cell type	Prokaryotic	(A)	Eukaryotic	Eukaryotic	Eukaryotic
(ii)	Cell wall	(B)	Present / absent	Present	Present	Absent
(iii)	Nuclear Membrane	Absent	Present	Present	(C)	Present
(iv)	Body organisation	Cellular	(D)	Multicellular	Tissue / organ	Tissue / Organ / Organ system

- (1) A-Eukaryotic, B-Noncellulosic, C-Present, D-Cellular
 (2) A-Prokaryotic, B-Cellulosic, C-Absent, D-Tissue
 (3) A-Eukaryotic, B-Present, C-Absent, D-Multicellular
 (4) A-Eukaryotic, B-Absent, C-May be present, D-Organ

24. Which is not true for the nucleus of prokaryotes-

- (1) The nucleus is indistinct and called as "nucleoid"
 (2) Nucleolus and nuclear membrane are not present in it
 (3) Non-histone proteins and histone proteins are not present in it.
 (4) It is also called as incipient nucleus or fibrillar nucleus

25. In a culture the initial concentration of bacterial cells is 2×10^5 per liter. If all bacterial cells divide every 20 minute, what will be the concentration after 1 hour and 30 minutes-

- (1) 16×10^5
 (2) 32×10^5
 (3) 2×10^{20}
 (4) 8×10^5

26. Which one of the following is mismatched?

- (1) Penicillium – Source of antibiotics
 (2) Albugo – Parasitic fungi on legume
 (3) Neurospora – used in biochemical and genetic work
 (4) Agaricus – edible fungus

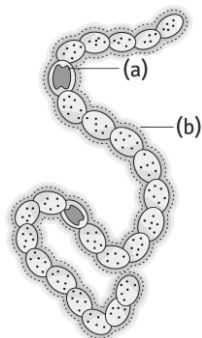
27. Which of the following pair is correctly matched?

- (1) Rhizobium - Parasite in the roots of leguminous plants
 (2) Mycorrhiza - Mineral uptake from soil
 (3) Yeast - Production of biogas
 (4) Myxomycetes - The disease Ring worm

28. In prokaryotes, chromatophores are :-

- (1) Specialised granules responsible for colouration of cells
 (2) Structure responsible for organising the shape of organism
 (3) Inclusion bodies lying free inside the cells for carrying out various metabolic activities
 (4) Internal membrane systems that may become extensive and complex in photosynthetic bacteria

29. In given diagram (a) and (b) represents to:

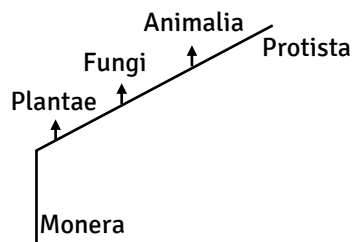


- (1) Heterocyst & mucilaginous sheath respectively
- (2) Mucilaginous sheath & heterocyst respectively
- (3) Harmogonia & mucilaginous sheath respectively
- (4) Heterocyst and trichome respectively

30. Read the following four statements (A – D):
 (A) In the five-kingdom classification, *Chlamydomonas* and *Chlorella* have been included in protista
 (B) *Entamoeba coli* a commonly occurring bacterium in human intestine
 (C) *Paramecium* and *Plasmodium* belong to the same kingdom as that of *Penicillium*
 (D) Hydrogen donor in photosynthesis of purple sulphur bacteria is H_2S
 How many of the above statements are incorrect?
 (1) Four (2) One
 (3) Two (4) Three

31. In which one of the following, the kingdom its two characters and its examples are not correctly matched, whereas the remaining three are correct
 (1) **Kingdom:** Protista; **Two characters:** a. Single celled, b. Eukaryotes; **Examples:** *Gonyaulax*, *Euglena*
 (2) **Kingdom:** Monera; **Two characters:** a. Single celled, b. prokaryotes; **Examples:** *Nostoc*, *Vibrio*
 (3) **Kingdom:** Mycota; **Two characters:** a. Multicellular, b. Chitinous cell wall; **Examples:** *Mucor*, *Aspergillus*
 (4) **Kingdom:** Animalia; **Two characters:** a. Eukaryotes, b. Cell wall absent **Examples:** *Amoeba*, *Plasmodium*

32. For above given cladogram select out the correct one :-



- (1) Animalia directly arose from monera
- (2) Fungi are the ancestor of Animalia
- (3) Plantae and protista arose from Monera directly
- (4) Plantae, fungi and Animalia arose from protista

33. Which of the following option is not related with protist and plantae but related with monera :

- (1) Cell wall and cell membrane
- (2) Ribosome and flagella
- (3) Nitrogen fixation ability
- (4) Cellular organization

34. Consider the following statements. Which of the statements given below is incorrect?

- (1) All prokaryotic organisms were grouped together under kingdom–Monera
- (2) The unicellular eukaryotic organisms were placed in kingdom–Protista
- (3) *Chlorella* and *Chlamydomonas*, both lack cell walls
- (4) *Paramecium* and *Amoeba* lack cell wall.

35. Choose the incorrect statement about members of kingdom–Monera.

- (1) Many of them live in or on other organisms as parasites
- (2) Some synthesise their own food from inorganic solutes
- (3) Bacterial structure is very complex though they have very simple behaviour
- (4) Hundreds of bacteria are present in handful of soil

36. Read the following statements about bacteria and select the correct option.
- (1) Bacteria are simple in structure, but complex in behaviour
 - (2) Bacteria are complex in structure, but simple in behaviour
 - (3) Bacteria are simple in both structure and behaviour
 - (4) Bacteria are complex in both structure and behaviour
37. Which of the following statements about methanogens is not correct?
- (1) They can be used to produce biogas
 - (2) They are found in the rumen of cattle and their excreta
 - (3) They grow aerobically and breakdown cellulose rich food
 - (4) They produce methane gas
38. Which of the following statements is/are incorrect?
- (1) Bacteria reproduce only by binary fission
 - (2) Under unfavourable conditions, bacteria produce several types of spores
 - (3) Bacteria reproduce by a sort of sexual reproduction by adopting a primitive type of RNA transfer from one bacterium to other
 - (4) Both (1) and (2)
39. Which one of the following statements is incorrect?
- (1) Golden algae are also called desmids
 - (2) Eubacteria are also called false bacteria
 - (3) Phycomycetes are also called algal fungi
 - (4) Cyanobacteria are also called blue-green algae
40. Which statement is correct?
- (1) Mycoplasma is smallest and wall less living organism
 - (2) Influenza and herpes are caused by virus having DNA and RNA
 - (3) *Nostoc* and *Anabaena* are important decomposers
 - (4) Methanogen are methane producing bacteria in wheat crops
41. Select the incorrect statement.
- (1) Bacterial cell wall is made up of peptidoglycan
 - (2) Pili and fimbriae are mainly involved in motility of bacterial cells
 - (3) Cyanobacteria lack flagellated cells
 - (4) Mycoplasma is a wall-less microorganism
42. Consider the following statements about mycoplasma. Which of the statement given below is incorrect?
- (1) They are pleomorphic bacteria, which lack cell wall
 - (2) Mycoplasma is the smallest living organism
 - (3) They cannot survive without oxygen
 - (4) Many mycoplasma are pathogenic in animals and plants
43. Select the incorrect statement.
- (1) The walls of diatoms are easily destructible
 - (2) 'Diatomaceous earth' is formed by the cell walls of diatoms
 - (3) Diatoms are chief producers in the oceans
 - (4) Diatoms are microscopic and float passively in water
44. Which of the following statement about *Euglena* is/are true?
- (1) Euglenoids bear flagella
 - (2) *Euglena* when placed in continuous darkness, lose their photosynthetic activity and die
 - (3) The pigments of *Euglena* are quite different from those of green plants
 - (4) *Euglena* is a marine protist
45. Which of the statement(s) given below is/are correct for amoeboid protozoans?
- (1) Live in freshwater, sea water or moist soil
 - (2) Has pseudopodia for locomotion and capturing prey
 - (3) Have silica shells on their surface in marine forms
 - (4) All of the above

46. Which of the following statements is correct?
(1) Slime moulds are haploid
(2) Protozoans lack cell wall
(3) Dinoflagellates are immotile
(4) Pellicle is absent in Euglena
47. Consider the following statements about Ascomycetes. Which one of the statement given below is false?
(1) They are saprophytic, decomposer, coprophilous and parasitic
(2) Include unicellular and multicellular forms
(3) Mycelium is coenocytic and aseptate
(4) Aspergillus, Claviceps and Neurospora are important examples of Ascomycetes
48. Which of the following statement is incorrect?
(1) Claviceps is a source of many alkaloids and LSD
(2) Conidia are produced exogenously and ascospores endogenously
(3) Yeasts have filamentous bodies with long thread-like hyphae
(4) Morels and truffles are edible delicacies
49. Which of the following statement is correct?
(1) Lichens do not grow in polluted areas
(2) Algal component of lichens is called mycobiont
(3) Fungal component of lichens is called phycobiont
(4) Lichens are not good pollution indicators
50. Which of the following statement(s) is/are incorrect?
(1) Protistan body includes a well-defined nucleus, and all cellular organelles
(2) Protists have nucleus along with flagella and cilia
(3) Protist cells have no nucleus but have some cellular organelles to perform basic functions
(4) All of the above
51. Which of the statement(s) given below is/are correct?
(1) Kingdom-Protista forms a link between monerans and the other organisms like plants, animal and fungi
(2) Protists reproduce asexually and sexually by a process involving cell fusion and zygote formation
(3) Being eukaryotes, the protistan cell body contains a well-defined nucleus and other membrane bound organelles
(4) All of the above
52. Which of the following statement about plant is false?
(1) Plants are heterotrophs
(2) Plants show alternation of generations during their life cycle
(3) Plants are multicellular eukaryotes
(4) Plants are non-motile
53. Select incorrect statement.
(1) The viroids were discovered by DJ Ivanowsky
(2) WM Stanley showed that viruses could be crystallised
(3) The term '*Contagium vivum fluidum*' was coined by MW Beijerinck
(4) Mosaic disease in tobacco and AIDS in human being are caused by viruses
54. State whether the given statements are true or false.
I. Bacteria show both autotrophic and heterotrophic nutrition.
II. Some of the bacteria are autotrophic. They may be photosynthetic autotrophic or chemosynthetic autotrophic.
III. Heterotrophic nutrition involves obtaining of readymade organic food from outside sources.
(1) I and II are true
(2) I is true, II and III are false
(3) I, II and III are true
(4) Only I is true

55. Read the following statements regarding archaebacteria and select the correct option.

- I. Archaebacteria differ from other bacteria in having different cell wall structure.
- II. Their cell wall is made up of cellulose and contains high amount of unsaturated fatty acid, which is responsible for their survival in extreme conditions.
- III. Thermoacidophiles have dual ability to tolerate high temperature as well as high acidity.

Which of the statements given above are correct?

- (1) I and II
- (2) I and III
- (3) II and III
- (4) All of the above

56. Analyze the following statements and identify the correct option given below.

- I. In diatoms the walls are embedded with silica and thus the walls are indestructible.
- II. Diatoms have left behind large amount of cell wall deposits in their habitat, this accumulation over billions of years is referred to as diatomaceous deposition or diatomaceous earth.

- (1) I is true, but II is false
- (2) I is false, but II is true
- (3) I and II are true
- (4) I and II are false

57. **Assertion** :- Lichen shows symbiotic relationship between algae and fungi.

Reason :- Algae absorb water and mineral nutrients from the environment and provide fungi.

- (1) both A and R true and R is the correct explanation of A
- (2) both A and R true but R is not correct explanation A
- (3) A is true but R is false
- (4) A is false and R is true

58. **Assertion:-** Although Bacterial structure is very simple yet they are very complex in behaviour.

Reason:- Bacteria as a group show most extensive metabolic diversity.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

59. **Assertion:-** Archaebacteria differ from other bacteria only in structure not, in resistance.

Reason:- Cell wall of Archaebacteria is made up of complex peptidoglycans instead of simple peptidoglycans.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

60. **Assertion:-** Heterocysts are specialised cells of blue green algae for Nitrogen fixation.

Reason:- In Heterocyst cells oxygen depleted conditions can be created, which is essential for Nitrogenase functioning.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

61. Assertion:- Chemosynthetic autotrophic bacteria play a great role in recycling nutrients like nitrogen, phosphorus, iron and sulphur.

Reason:- Chemosynthetic autotrophic bacteria oxidise various inorganic substances to obtain energy

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

62. Assertion:- Members of protista are eukaryotes.

Reason:- Protistial cell body does not contain a well defined nucleus and other membrane bounded cell organelles.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

63. Assertion:- The cell wall of diatoms is indestructible.

Reason:- The cell wall of diatoms is embedded with calcium carbonates.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

64. Assertion:- Diatoms are the chief producers in the oceans.

Reason:- Diatoms possess both chlorophyll-a and chlorophyll-c.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

65. Assertion:- Euglenoids can perform photosynthesis.

Reason:- They have chloroplast in cytoplasm.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

66. Assertion:- Spore is the exclusive structure of slime moulds upon which cell wall remain present.

Reason:- Slime moulds differentiate and form fruiting body during unfavourable conditions in which spores remain filled.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

67. Assertion:- Fungi prefer to grow in cold and humid areas.

Reason:- Low temperature favours the metabolic activities.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

68. Assertion:- In Ascomycetes and Basidiomycetes well distinct Dikaryophase present during sexual life.

Reason:- During sexual reproduction of Ascomycetes and Basidiomycetes plasmogamy is immediately followed by karyogamy.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

69. Assertion:- Neurospora fungus is known as Drosophila of plant kingdom.

Reason:- Like Drosophila, Neurospora is also extensively used in biochemical and genetic work.

- (1) If both assertion and reason are true and reason is the correct explanation of

assertion.

- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.

(3) If assertion is true but reason is false.

(4) If both assertion and reason are false.

70. Assertion : Archaeobacteria are most resistant to adverse environmental conditions.

Reason: In archaeobacterial, cell wall and cell membrane are highly complex.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

71. Assertion:- According to Whittaker's classification dinoflagellates are included in kingdom Protista.

Reason:- Dinoflagellates have autotrophic type of nutrition.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

Exercise 4 (Previous Year's Questions)

1. Single-celled eukaryotes are included in :-
[AIPMT – 2010]
(1) Monera (2) Protista
(3) Fungi (4) Archaea
2. Ringworm in humans is caused by:
[AIPMT – 2010]
(1) Viruses (2) Bacteria
(3) Fungi (4) Nematodes
3. The common nitrogen-fixer in paddy fields is
[AIPMT – 2010]
(1) Frankia (2) Rhizobium
(3) Azospirillum (4) Oscillatoria
4. Some hyperthermophilic organisms that grow in highly acidic (pH2) habitats belong to the two groups :-
[AIPMT – 2010]
(1) Liverwort and Yeast
(2) Eubacteria and Archaea
(3) Cyanobacteria and Diatoms
(4) Protists and mosses
5. Black (stem) rust of wheat is caused by-
[AIPMT – 2010]
(1) Ustilago nuda
(2) Puccinia graminis
(3) Xanthomonas oryzae
(4) Alternaria solani
6. One of the free-living, anaerobic nitrogen-fixer is :
[AIPMT – 2010]
(1) Azotobacter (2) Beijernickia
(3) Rhodospirillum (4) Rhizobium
7. Membrane-bound organelles are absent in :
[AIPMT – 2010]
(1) Plasmodium (2) Saccharomyces
(3) Streptococcus (4) Chlamydononas
8. Select the correct combination of the statements (a-d) regarding the characteristics of certain organisms
[AIPMT – 2010]
(a) Methanogens are Archaeobacteria which produce methane in marshy areas.
(b) Nostoc is a filamentous blue-green alga which fixes atmospheric nitrogen.
(c) Chemosynthetic autotrophic bacteria synthesize cellulose from glucose
(d) Mycoplasma lack a cell wall and can survive with oxygen
The correct Statements are:
(1) (a), (b), (c) (2) (b), (c), (d)
(3) (a), (b), (d) (4) (b), (c)
9. Ethanol is commercially produced through a particular species of :
[AIPMT – 2011]
(1) Saccharomyces (2) Clostridium
(3) Trichoderma (4) Aspergillus
10. An organism used as a bio-fertilizer for raising soyabean crops is :-
[AIPMT – 2011]
(1) Azotobacter (2) Azospirillum
(3) Rhizobium (4) Nostoc
11. Nitrifying bacteria :-
[AIPMT – 2011]
(1) Oxidize ammonia to nitrates
(2) Convert free nitrogen to nitrogen compounds
(3) Convert proteins into ammonia
(4) Reduce nitrates to free nitrogen
12. Which one of the following is a wrong matching of a microbe and its industrial product, while the remaining three are correct ?
[AIPMT – 2011]
(1) Aspergillus niger -citric acid
(2) Yeast -Statins
(3) Acetobacter aceti-acetic acid
(4) Clostridium butylicum-lactic acid
13. Which one of the following is not a biofertilizer?
[AIPMT – 2011]
(1) Agrobacterium (2) Rhizobium
(3) Nostoc (4) Mycorrhiza
14. The pathogen Microsporium responsible for ringworm disease in humans belongs to the same Kingdom of organisms as that of-
[AIPMT – 2011]
(1) Ascaris, a round worm
(2) Taenia, a tapeworm
(3) Wuchereria, a filarial worm
(4) Rhizopus, a mould
15. The cyanobacteria are also referred to as :
[AIPMT – 2012]
(1) Slime moulds (2) Blue green algae
(3) Protists (4) Golden algae
16. Yeast is used in the production of :-
[AIPMT – 2012]
(1) Bread and beer
(2) Cheese and butter
(3) Citric acid and lactic acid
(4) Lipase and pectinase

- 17.** A nitrogen-fixing microbe associated with Azolla in rice fields is- **[AIPMT – 2012]**
 (1) Frankia (2) Tolypothrix
 (3) Spirulina (4) Anabaena
- 18.** Which one of the following does not differ in E.coli and Chlamydomonas? **[AIPMT – 2012]**
 (1) Cell wall
 (2) Cell membrane
 (3) Ribosomes
 (4) Chromosomal Organization
- 19.** Maximum nutritional diversity is found in the group- **[AIPMT – 2012]**
 (1) Monera (2) Plantae
 (3) Fungi (4) Animalia
- 20.** Nuclear membrane is absent in :- **[AIPMT – 2012]**
 (1) Volvox (2) Nostoc
 (3) Penicillium (4) Agaricus
- 21.** The most abundant prokaryotes helpful to humans in making curd from milk and in the production of antibiotics are the ones categorised as :- **[AIPMT – 2012]**
 (1) Chemosynthetic autotrophs
 (2) Heterotrophic bacteria
 (3) Cyanobacteria
 (4) Archaeobacteria
- 22.** In the five-kingdom classification, Chlamydomonas and Chlorella have been included in : **[AIPMT – 2012]**
 (1) Plantae (2) Monera
 (3) Protista (4) Algae
- 23.** How many organisms in the list given below are autotrophs? **[AIPMT – 2012]**
 Lactobacillus, Nostoc, Chara, Nitrosomonas, Nitrobacter, Streptomyces, Saccharomyces, Trypanosoma, Porphyra, Wolfia
 (1) Six (2) Three
 (3) Four (4) Five
- 24.** Which one of the following is the wrong statement? **[AIPMT – 2012]**
 (1) Phosphorus is a constituent of cell membranes, certain nucleic acids and all proteins
 (2) Nitrosomonas and Nitrobacter are chemoautotrophs
 (3) Anabaena and Nostoc are capable of fixing nitrogen in free-living state also
 (4) Root nodule forming nitrogen fixers live as aerobes under free-living conditions
- 25.** Which one single organism or the pair of organisms is correctly assigned to its or their named taxonomic group? **[AIPMT – 2012]**
 (1) Yeast used in making bread and beer is a fungus
 (2) Nostoc and Anabaena are examples of protista
 (3) Paramecium and Plasmodium belong to the same kingdom as that of Penicillium
 (4) Lichen is a composite organism formed from the symbiotic association of an algae and a protozoan
- 26.** Which of the following are likely to be present in deep sea water ? **[NEET UG – 2013]**
 (1) Saprophytic fungi (2) Archaeobacteria
 (3) Eubacteria (4) Blue-green algae
- 27.** A good producer of citric acid is- **[NEET UG – 2013]**
 (1) Saccharomyces (2) Aspergillus
 (3) Pseudomonas (4) Clostridium
- 28.** Select the wrong statement: **[NEET UG – 2013]**
 (1) Chlamydomonas exhibits both isogamy and anisogamy and fucus shows oogamy
 (2) Isogametes are similar in structure, function and behaviour
 (3) Anisogametes differ either in structure, function or behaviour
 (4) In Oomycetes female gamete is smaller and motile, while male gamete is larger and nonmotile

29. Which of the following represents the maximum number of species among global diversity? **[NEET UG – 2013]**
 (1) Mosses and fern (2) Algae
 (3) Lichens (4) Fungi
30. Pigments containing membranous extensions in some cyanobacteria are: **[NEET UG – 2013]**
 (1) Chromatophores
 (2) Heterocysts
 (3) Basal bodies
 (4) Pneumatophores
31. Which of the following is not correctly matched for the organism and its cell wall degrading enzyme- **[NEET UG – 2013]**
 (1) Fungi-Chitinase
 (2) Bacteria-Lysozyme
 (3) Plant cell-Cellulase
 (4) Algae-Methylase
32. Five kingdom system of classification suggested by R.H. Whittaker is not based on : **[AIPMT – 2014]**
 (1) Presence or absence of a well defined nucleus
 (2) Mode of reproduction
 (3) Mode of nutrition
 (4) Complexity of body organization
33. Which one of the following fungi contains hallucinogens? **[AIPMT – 2014]**
 (1) Morchella esculenta
 (2) Amanita muscaria
 (3) Neurospora sp.
 (4) Ustilago sp.
34. Archaeobacteria differ from eubacteria in: **[AIPMT-2014]**
 (1) Cell membrane
 (2) Mode of nutrition
 (3) Cell shape
 (4) Mode of reproduction
35. Anoxygenic photosynthesis is characteristic of : **[AIPMT – 2014]**
 (1) Rhodospirillum (2) Spirogyra
 (3) Chlamydomonas (4) Ulva
36. Which of the following shows coiled RNA strands and capsomeres? **[AIPMT – 2014]**
 (1) Polio virus
 (2) Tobacco mosaic virus
 (3) Measles virus
 (4) Retrovirus
37. Transformation was discovered by: **[AIPMT – 2014]**
 (1) Meselson and Stahl
 (2) Hershey and Chase
 (3) Griffith
 (4) Watson and Crick
38. Viruses have: **[AIPMT – 2014]**
 (1) DNA enclosed in a protein coat
 (2) Prokaryotic nucleus
 (3) Single chromosome
 (4) Both DNA and RNA
39. Which one of the following living organisms completely lacks a cell wall? **[AIPMT – 2014]**
 (1) Cyanobacteria (2) Seafan(Gorgonia)
 (3) Saccharomyces (4) Blue-green algae
40. Which one of the following matches is correct? **[AIPMT – 2015]**
 (1) a-Alternaria, b-Sexual reproduction absent, c-Deuteromycetes
 (2) a-Mucor, b-Reproduction by Conjugation, c-Ascomycetes
 (3) a-Agaricus, b-Parasitic fungus, c-Basidiomycetes
 (4) a-Phytophthora, b-Aseptate mycelium, c-Basidiomycetes
41. True nucleus is absent in : **[AIPMT – 2015]**
 (1) Mucor (2) Volvox
 (3) Vaucheria (4) Anabaena
42. Which one of the following is not an inclusion body found in prokaryotes? **[AIPMT – 2015]**
 (1) Cyanophyccean granule
 (2) Glycogen granule
 (3) Polysome
 (4) Phosphate granule

43. The guts of cow and buffalo possess :-
[AIPMT – 2015]
(1) Chlorella spp. (2) Methanogens
(3) Cyanobacteria (4) Fucus spp.
44. Cell wall is absent in - [RE AIPMT – 2015]
(1) Nostoc (2) Aspergillus
(3) Funaria (4) Mycoplasma
45. Chromatophores take part in ?
[RE AIPMT – 2015]
(1) Respiration (2) Photosynthesis
(3) Growth (4) Movement
46. The structures that help some bacteria to attach to rocks and / or host tissues are:
[RE AIPMT – 2015]
(1) Holdfast (2) Rhizoids
(3) Fimbriae (4) Mesosomes
47. Which of the following are not membrane bound ?
[RE AIPMT-2015]
(1) Mesosomes
(2) Vacuoles
(3) Ribosomes
(4) Lysosomes
48. Which of the following structures is not found in prokaryotic cells ?
[RE AIPMT – 2015]
(1) Plasma membrane
(2) Nuclear envelope
(3) Ribosome
(4) Mesosome
49. The imperfect fungi which are decomposer of litter and help in mineral cycling belong to :
[RE AIPMT – 2015]
(1) Ascomycetes (2) Deuteromycetes
(3) Basidiomycetes (4) Phycomycetes
50. Pick up the wrong statement:
[RE AIPMT – 2015]
(1) Nuclear membrane is present in Monera
(2) Cell wall is absent in Animalia
(3) Protista have photosynthetic and heterotrophic modes of nutrition
(4) Some fungi are edible

51. Choose the wrong statement :
[RE AIPMT – 2015]
(1) Yeast is unicellular and useful in fermentation
(2) Penicillium is multicellular and produces antibiotics
(3) Neurospora is used in the study of biochemical genetics
(4) Morels and truffles are poisonous mushrooms
52. In which group of organisms the cell walls form two thin overlapping shells which fit together?
[RE AIPMT – 2015]
(1) Slime moulds (2) Chrysophytes
(3) Euglenoids (4) Dinoflagellates
53. Select the wrong statements:
[NEET-II – 2016]
(1) Diatomaceous earth is formed by the cell walls of diatoms
(2) Diatoms are chief producers in the oceans
(3) Diatoms are microscopic and float passively in water
(4) The wall of diatoms are easily destructible
54. Methanogens belong to : [NEET-II – 2016]
(1) Archaeobacteria (2) Dinoflagellates
(3) Slime moulds (4) Eubacteria
55. Chrysophytes, Euglenoids, Dinoflagellates and Slime moulds are included in the kingdom:
[NEET-I – 2016]
(1) Animalia (2) Protista
(3) Monera (4) Fungi
56. One of the major components of cell wall of most fungi is:
[NEET-I – 2016]
(1) Chitin (2) Peptidoglycan
(3) Cellulose (4) Hemicellulose
57. Which one of the following statements is wrong ?
[NEET-I – 2016]
(1) Cyanobacteria are also called blue-green algae
(2) Golden algae are also called desmids.
(3) Eubacteria are also called false bacteria.
(4) Phycomycetes are also called algal fungi.

58. Which of the following statements is wrong for viroids? **[NEET-I – 2016]**
 (1) They lack a protein coat
 (2) They are smaller than viruses
 (3) They cause infections
 (4) Their RNA is of high molecular weight
59. Which of the following sets of diseases is caused by bacteria? **[NEET-II – 2016]**
 (1) Typhoid and smallpox
 (2) Tetanus and mumps
 (3) Herpes and influenza
 (4) Cholera and tetanus
60. Select the wrong statement- **[NEET-II – 2016]**
 (1) Pili and fimbriae are mainly involved in motility of bacterial cells.
 (2) Cyanobacteria lack flagellated cells
 (3) Mycoplasma is a wall less microorganism
 (4) Bacterial cell wall is made up of peptidoglycan
61. Which one of the following is wrong for fungi? **[NEET-II – 2016]**
 (1) All fungi possess a purely cellulosic cell wall
 (2) They are heterotrophic
 (3) They are both unicellular and multicellular
 (4) They are eukaryotic
62. Which of the following is correctly matched for the product produced by them? **[NEET – 2017]**
 (1) *Acetobacter aceti* : Antibiotics
 (2) *Methanobacterium* : Lactic acid
 (3) *Penicillium notatum* : Acetic acid
 (4) *Saccharomyces cerevisiae* : Ethanol
63. Mycorrhizae are the example of: **[NEET – 2017]**
 (1) Fungistasis (2) Amensalism
 (3) Antibiosis (4) Mutualism
64. Which of the following are found in extreme saline conditions? **[NEET – 2017]**
 (1) Archaeobacteria (2) Eubacteria
 (3) Cyanobacteria (4) Mycobacteria
65. Which among the following are the smallest living cells, known without a definite cell wall, pathogenic to plants as well as animals and can survive without oxygen? **[NEET – 2017]**
 (1) *Bacillus* (2) *Pseudomonas*
 (3) *Mycoplasma* (4) *Nostoc*
66. Select the mismatch: **[NEET – 2017]**
 (1) *Frankia* - *Alnus*
 (2) *Rhodospirillum* - Mycorrhiza
 (3) *Anabaena* - Nitrogen fixer
 (4) *Rhizobium* - Alfalfa
67. Which of the following organisms are known as chief producers in the oceans? **[NEET – 2018]**
 (1) Dinoflagellates (2) Diatoms
 (3) Cyanobacteria (4) Euglenoids
68. Oxygen is not produced during photosynthesis by: **[NEET – 2018]**
 (1) Green sulphur bacteria
 (2) *Nostoc*
 (3) *Cycas*
 (4) *Chara*
69. Which among the following is not a prokaryote? **[NEET – 2018]**
 (1) *Saccharomyces* (2) *Mycobacterium*
 (3) *Nostoc* (4) *Oscillatoria*
70. After karyogamy followed by meiosis, spores produced exogenously in: **[NEET – 2018]**
 (1) *Neurospora* (2) *Alternaria*
 (3) *Agaricus* (4) *Saccharomyces*
71. Select the wrong statement. **[NEET – 2018]**
 (1) Cell wall is present in members of fungi and plantae.
 (2) Mushroom belong to basidiomycetes
 (3) Pseudopodia are locomotory and feeding structures in sporozoans
 (4) Mitochondria are the powerhouse of the cell in all kingdom except monera

72. Ciliates differ from all other protozoans in-
[NEET – 2018]

- (1) Using flagella for locomotion
- (2) Having a contractile vacuole for removing excess water
- (3) Using pseudopodia for capturing prey
- (4) Having two types of nuclei

73. Match **Column-I** with **column-II**.

	Column-I		Column-II
A.	Saprophyte	i.	Symbiotic association of fungi with plant roots
B.	Parasite	ii.	Decomposition of dead organic materials
C.	Lichens	iii.	Living on living plants or animals
D.	Mycorrhiza	iv.	Symbiotic association of algae and fungi

Choose the **correct** answer from the options given below [NEET – 2019]

- (1) A=ii, B= i, C = iii, D=iv
- (2) A=ii, B= iii, C = iv, D=i
- (3) A=i, B= ii, C = iv, D=i
- (4) A=iii, B= ii, C = i, D=iv

74. Thiobacillus is a group of bacteria helpful in carrying out: [NEET – 2019]

- (1) Nitrification
- (2) Denitrification
- (3) Nitrogen fixation
- (4) Chemoautotrophic fixation

75. Which of the following statements is **incorrect**? [NEET – 2019]

- (1) Infective constituent in viruses is the protein coat.
- (2) Prions consist of abnormally folded proteins
- (3) Viroids lack a protein coat.
- (4) Viruses are obligate parasites.

76. Which of the following statements is **incorrect**? [NEET – 2019]

- (1) Conidia are produced exogenously and ascospores endogenously
- (2) Yeasts have filamentous bodies with long thread like hyphae
- (3) Morels and truffles are edible delicacies
- (4) Claviceps is a source of many alkaloids and LSD.

77. According to Robert May, the global species diversity is about- [NEET-I – 2020]

- (1) 20 million
- (2) 50 million
- (3) 7 million
- (4) 1.5 million

78. Which of the following is correct about viroids? [NEET-I – 2020]

- (1) They have free RNA without protein coat.
- (2) They have DNA with protein coat.
- (3) They have free DNA without protein coat.
- (4) They have RNA with protein coat.

79. Match **column-I** with **column-II**.

[NEET – 2021]

	Column-I		Column-II
(a)	Nitrococcus	(i)	Denitrification
(b)	Rhizobium	(ii)	Conversion of ammonia to nitrate
(c)	Thiobacillus	(iii)	Conservation of nitrite to nitrate
(d)	Nitrobacter	(iv)	Conversion of atmospheric nitrogen to ammonia

Choose the **correct** answer from options given below.

- (1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
- (2) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
- (3) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
- (4) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

80. Which of the following statements is **correct**? [NEET – 2021]

- (1) Some of the organisms can fix atmospheric nitrogen in specialized cells called sheath cells.
- (2) Fusion of two cells is called Karyogamy.
- (3) Fusion of protoplasts between two motile or non-motile gametes is called Plasmogamy.
- (4) Organisms that depend on living plants are called saprophytes.

81. Given below are two statements :-
Statement I : Mycoplasma can pass through less than 1 micron filter size.
Statement II : Mycoplasma are bacteria with cell wall
 In the light of the above statements, choose the most appropriate answer from the options given below: **[NEET – 2022]**
- (1) Both Statement I and Statement II are incorrect
 - (2) Statement I is correct but Statement II is incorrect
 - (3) Statement I is incorrect but Statement II is correct
 - (4) both Statement I and Statement II are correct
82. Which of the following is a **correct** statement? **[NEET – 2022]**
- (1) Bacteria are exclusively heterotrophic organisms
 - (2) Slime moulds are saprophytic organisms classified under Kingdom Monera.
 - (3) Mycoplasma have DNA, Ribosome and cell wall
 - (4) Cyanobacteria are a group of autotrophic organisms classified under kingdom Monera.

83. Which one of the following is not a criterion for classification of fungi? **[NEET–2024]**
- (1) Mode of spore formation
 - (2) Fruiting body
 - (3) Morphology of mycelium
 - (4) Mode of nutrition

84. Match List I with List II – **[NEET–2024]**

	List-I		List-II
A.	Rhizopus	I.	Mushroom
B.	Ustilago	II.	Smut fungus
C.	Puccinia	III.	Bread mould
D.	Agaricus	IV.	Rust fungus

Choose the correct answer from the options given below :

- (1) A-III, B-II, C-I, D-IV
- (2) A-IV, B-III, C-II, D-I
- (3) A-III, B-II, C-IV, D-I
- (4) A-I, B-III, C-II, D-IV

ANSWER KEYS

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	11										
Ans.	2	3	4	1	1	2	4	2	1	4	3										

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14										
Ans.	2	3	4	1	2	1	2	3	3	3	4	1	3	4										

Exercise 1.3

Que.	1	2	3	4	5	6	7	8	9	10	11	12										
Ans.	4	1	2	1	1	3	3	3	3	4	2	3										

Exercise 1.4

Que.	1	2	3	4	5																	
Ans.	1	3	1	2	1																	

Exercise 1.5

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19		
Ans.	1	1	4	4	4	4	1	1	4	1	2	1	1	1	1	3	1	2	4		

Exercise 1.6

Que.	1	2	3	4																	
Ans.	3	4	2	1																	

Exercise 1.7

Que.	1	2	3	4	5	6	7															
Ans.	2	1	1	2	2	4	2															

Exercise 1.8

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	1	4	1	4	1	1	4	2	3	3	4	1	3	3	2	3	2	2	1
Que.	21	22																		
Ans.	2	1																		

Exercise 1.9

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	1	1	1	4	2	1	4	2	1	2	3	4	2	4	1	2	1	1	4
Que.	21	22	23	24																
Ans.	3	4	3	3																

Exercise 1.10

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	1	2	4	1	3	4	3	2	3	3	2	4	3	4	2	1	1	1	3
Que.	21	22	23	24																
Ans.	2	2	2	4																

Exercise 1.11

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	
Ans.	1	1	3	4	1	4	4	4	2	4	2	4	3	3	1	1	2	

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	3	3	3	2	1	1	1	2	1	1	4	1	1	3	3	1	4	4	4
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	4	4	1	3	3	4	3	3	4	2	4	4	2	2	2	3	4	4	3	4

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	3	1	2	2	4	3	3	1	1	4	2	3	2	2	2	2	2	3	3
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	2	2	1	3	2	2	2	4	1	3	4	4	3	3	3	1	3	2	2	1
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	2	3	1	1	4	2	3	3	1	3	4	1	1	3	2	3	3	1	4	1
Que.	61	62	63	64	65	66	67	68	69	70	71									
Ans.	1	3	3	2	1	1	4	3	1	1	2									

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	3	4	2	2	3	3	3	1	3	1	4	1	4	2	1	4	2	1	2
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	2	3	1	1	1	2	2	4	2	1	4	2	2	1	1	2	3	1	2	1
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	4	4	2	4	2	3	3	2	2	1	4	2	4	1	2	1	3	4	4	1
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
Ans.	1	4	4	1	3	2	2	1	1	3	3	4	2	2	1	2	3	1	2	3
Que.	81	82	83	84																
Ans.	2	4	4	3																